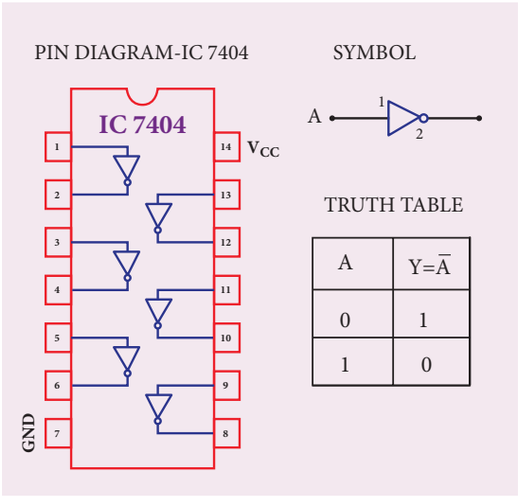
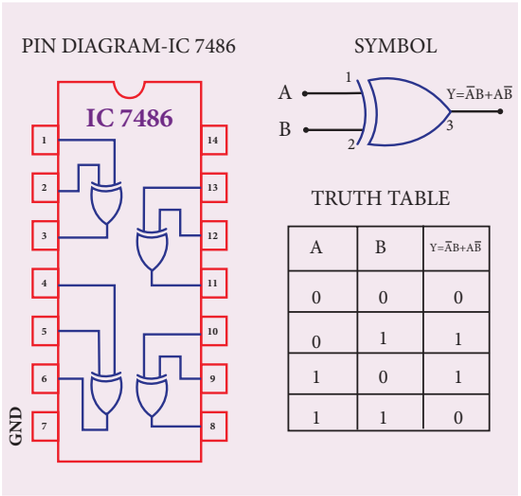
NOT Gate:



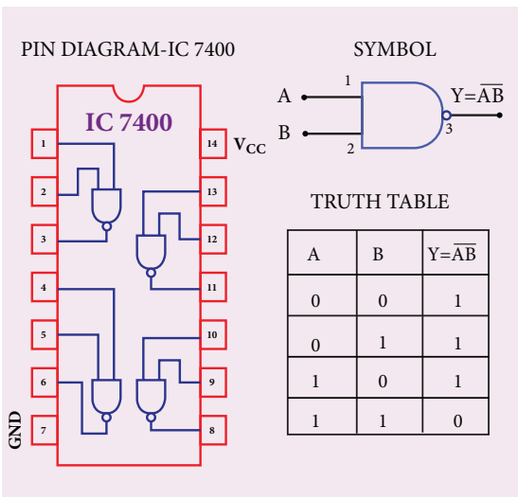
A	$Y=\overline{A}$
0	1
1	0

X-OR Gate :



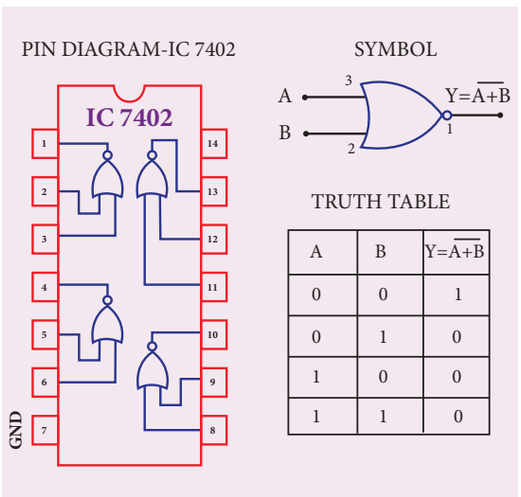
A	B	$Y=\overline{A}B+A\overline{B}$
0	0	0
0	1	1
1	0	1
1	1	0

NAND Gate:



A	B	$Y=\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate:



A	B	$Y=\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

PROCEDURE

- To verify the truth table of a logic gate, the suitable IC is taken and the connections are given using the circuit diagram.
- For all the ICs, 5V is applied to the pin 14 while the pin 7 is connected to the ground.
- The logical inputs of the truth table are applied and the corresponding output is noted.
- Similarly the output is noted for all other combinations of inputs.
- In this way, the truth table of a logic gate is verified.

RESULT

The truth table of logic gates AND, OR, NOT, Ex-OR, NAND and NOR using integrated circuits is verified.

Precautions

(i) V_{CC} and ground pins must not be interchanged while making connections. Otherwise the chip will be damaged. (ii) The pin configuration for NOR gate is different from other gates

XII-Physics_Vol-1) PRACTICAL.indd 315 08-12-2022 11:00:44